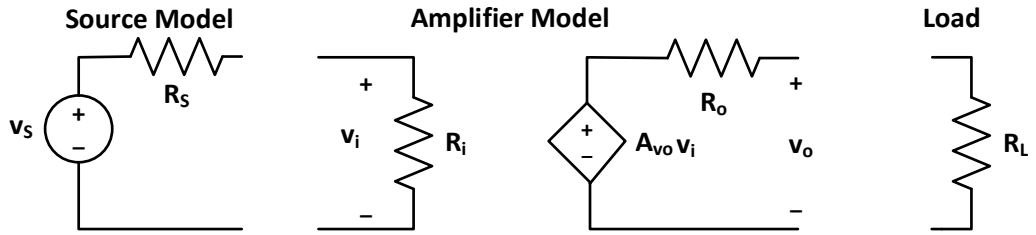
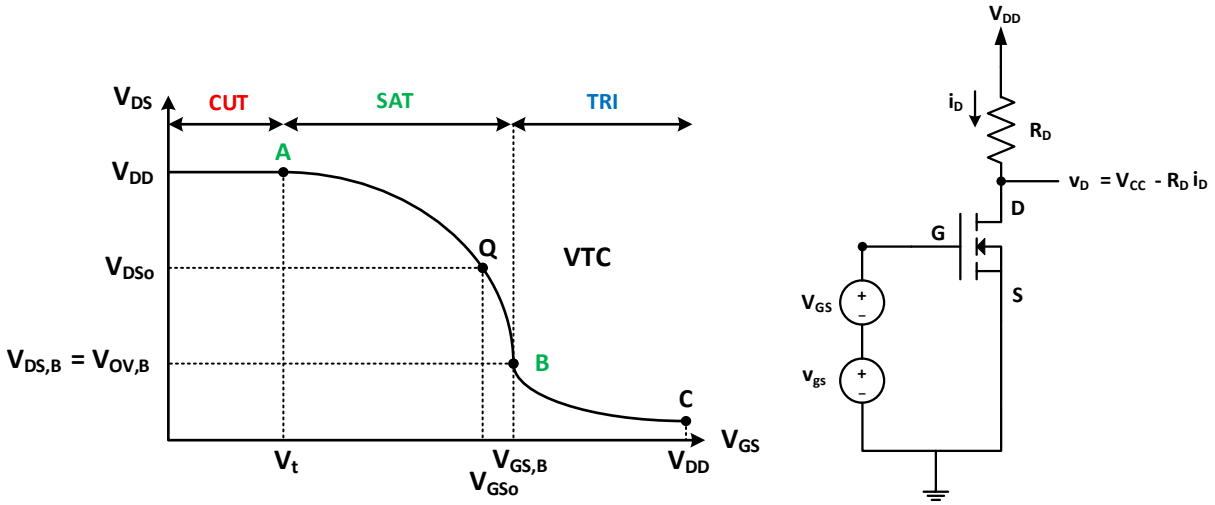
Equation Sheet: Transistor Amplifiers



$$G_V = \frac{v_o}{v_s} = \frac{R_i}{R_i + R_S} A_{vo} \frac{R_L}{R_L + R_o}$$

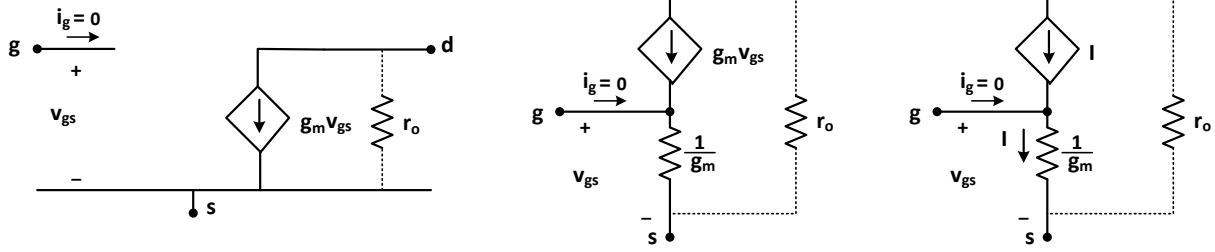
Common Source Amplifier Voltage Transfer Curve



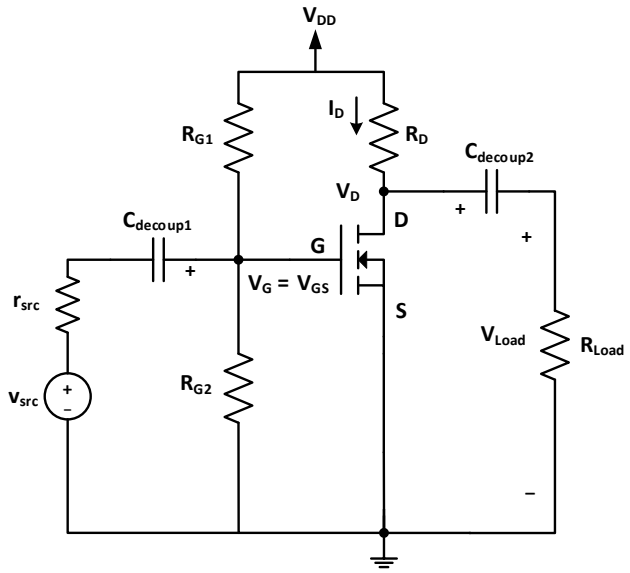
$$V_{DS,B} = \frac{\sqrt{1 + 2K_n R_D V_{DD}} - 1}{K_n R_D} = V_{ov,B} \qquad V_{GS,B} = V_{DS,B} + V_t \qquad g_m = K_n V_{OV} = \frac{2I_D}{V_{OV}} = \sqrt{2K_n I_D}$$

$$A_{vo} = -g_m R_D = -K_n V_{ov} R_D \qquad \left|A_{v,\max}\right| = \frac{V_{DD} - V_{OV,B}}{\frac{1}{2} V_{OV,B}} \qquad r_o \approx \frac{V_A}{I_D}$$

Small-Signal (AC) NMOS Transistor Models: Validity Check: $v_{gs} \ll 2V_{OV} \rightarrow v_{gs} < 0.2 \times V_{OV}$



CS-amplifier Design No R_S



STEP 1: Choose $\mathrm{V}_{DS} = \dfrac{\mathrm{V}_{DD}}{2}$

STEP 2: Compute $\mathrm{V}_{OV} = \dfrac{2\left(\mathrm{V}_{DD} - \mathrm{V}_{DS}\right)}{\left|\mathrm{A}_{vo}\right|}$ [Make sure this is not too small – constraining linearity]

STEP 3: Compute $\mathrm{R}_D = \dfrac{\left|\mathrm{A}_{vo}\right|}{\mathrm{V}_{OV}\ \mathrm{K}_n}$

STEP 4: Compute $\mathrm{V}_{GS} = \mathrm{V}_{OV} + \mathrm{V}_t$

STEP 5: Compute R_{G1} and R_{G2}: $\mathrm{R}_{G1} \geq \dfrac{\mathrm{V}_{DD}}{\mathrm{V}_{GS}} \times \mathrm{R}_{IN}$ $\mathrm{R}_{G2} = \dfrac{\mathrm{R}_{G1}}{\left(\dfrac{\mathrm{V}_{DD}}{\mathrm{V}_{GS}} - 1\right)}$

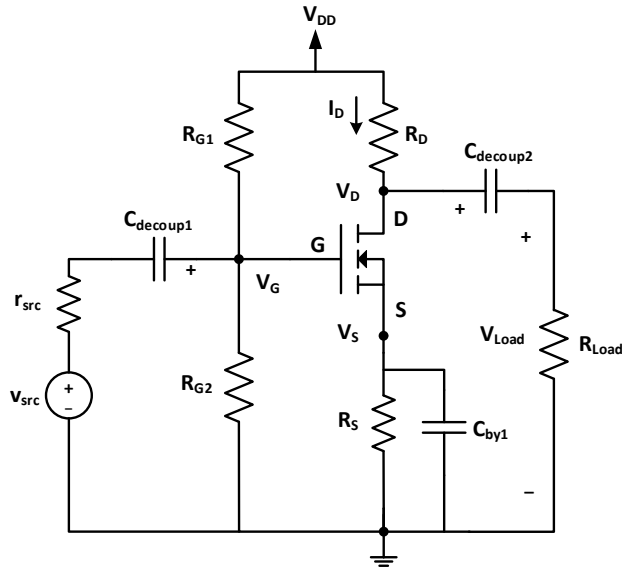
STEP 6: Check headroom and legroom: $v_{ds,pk} = \left|\mathrm{A}_{vo}\right| \times v_{gs,pk}$

$$v_{DS,\max} = \mathrm{V}_{DS} + v_{ds,pk} \qquad v_{DS,\min} = \mathrm{V}_{DS} - v_{ds,pk}$$

$$\mathrm{V}_{DS,B} = \frac{\sqrt{1 + 2\,\mathrm{K}_n\,\mathrm{R}_D\,\mathrm{V}_{DD}} - 1}{\mathrm{K}_n\,\mathrm{R}_D} = \mathrm{V}_{ov,B}$$

STEP 7: Check linearity: $v_{gs,pk} \leq 0.2 \times \mathrm{V}_{OV}$

CS-amplifier Design Fully AC-Bypassed R_S



STEP 1: Choose $\mathrm{V}_{RD} = \mathrm{V}_{DS} = \mathrm{V}_{RS} = \dfrac{\mathrm{V}_{DD}}{3}$

STEP 2: Compute $\mathrm{I}_D = 2 \times K_n \times \left[\dfrac{\mathrm{V}_{DD}}{3 \times \left| \mathrm{A}_{vo} \right|} \right]^2$

STEP 3: Compute $\mathrm{R}_D = \dfrac{\mathrm{V}_{RD}}{\mathrm{I}_D} = \mathrm{R}_S$

STEP 4: Compute $\mathrm{V}_{OV} = \sqrt{\dfrac{2 \times \mathrm{I}_D}{\mathrm{K}_n}}$ [Make sure this is not too small – constraining linearity]

STEP 5: Compute $\mathrm{V_G}$: $\mathrm{V}_{GS} = \mathrm{V}_{OV} + \mathrm{V}_t$ $\qquad$ $\mathrm{V}_G = \mathrm{V}_{GS} + \mathrm{V}_S$

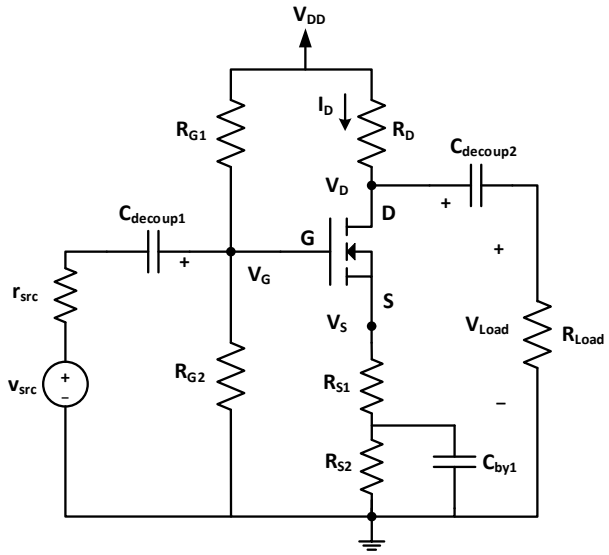
STEP 6: Compute $\mathrm{R_{G1}}$ and $\mathrm{R_{G2}}$: $\mathrm{R}_{G1} \geq \dfrac{\mathrm{V}_{DD}}{\mathrm{V}_G} \times \mathrm{R}_{IN}$ $\qquad$ $\mathrm{R}_{G2} = \dfrac{\mathrm{R}_{G1}}{\left(\dfrac{\mathrm{V}_{DD}}{\mathrm{V}_G} - 1 \right)}$

STEP 7: Check headroom and legroom: $v_{d,pk} = \left| \mathrm{A}_{vo} \right| \times v_{g,pk}$

$v_{D,\max} = \mathrm{V}_D + v_{d,pk}$ $\qquad$ $v_{D,\min} = \mathrm{V}_D - v_{d,pk}$ $\qquad$ $\mathrm{V}_D = \mathrm{V}_{DD} - \mathrm{V}_{RD}$

STEP 8: Check linearity: $v_{gs,pk} \leq 0.2 \times \mathrm{V}_{OV}$

CS-amplifier Design Partially AC-Bypassed R_S



STEP 1: Choose $V_{RD} = V_{DS} = V_{RS} = \frac{V_{DD}}{3}$; Choose $R_{S1} = \frac{1}{g_m}$; Choose $|A_{vo}|$ slightly above midrange

STEP 2: Compute $I_D = 2 \times K_n \times \left[\frac{V_{DD}}{6 \times |A_{vo}|} \right]^2$ [6 from $V_{DD}/3$ and $g_m R_D/2$ from $R_{S1} = 1/g_m$]

STEP 3: Compute $R_D = R_S = \frac{V_{DD}/3}{I_D}$

STEP 4: Compute $V_{OV} = \sqrt{\frac{2 \times I_D}{K_n}}$ [Make sure this is not too small – constraining linearity]

STEP 5: Compute $g_m = K_n \times V_{OV}$ then $R_{s1} = \frac{1}{g_m}$ $R_{s2} = R_S - R_{s1}$

STEP 6: Compute V_G: $V_{GS} = V_{OV} + V_t$ $V_G = V_{GS} + V_S$

STEP 7: Compute R_{G1} and R_{G2}: $R_{G1} \geq \frac{V_{DD}}{V_G} \times R_{IN}$ $R_{G2} = \frac{R_{G1}}{\left(\frac{V_{DD}}{V_G} - 1 \right)}$

STEP 8: Check headroom and legroom: $v_{d,pk} = |A_{vo}| \times v_{g,pk}$

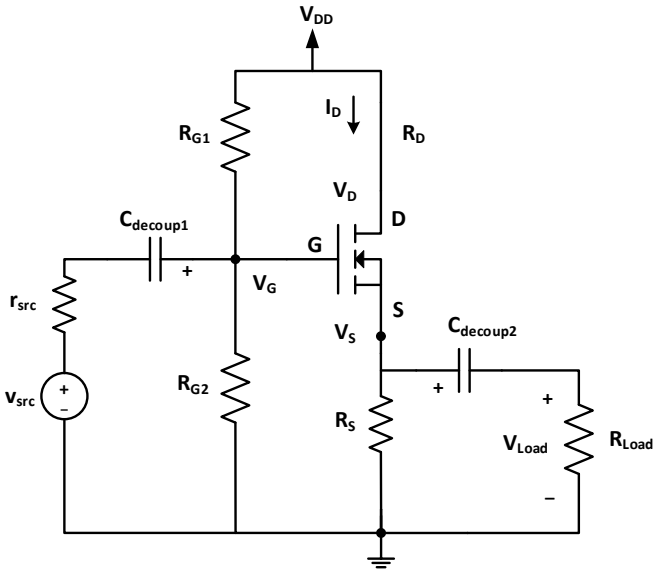
$v_{D,\max} = V_D + v_{d,pk}$ $v_{D,\min} = V_D - v_{d,pk}$ $V_D = V_{DD} - V_{RD}$

STEP 9: Check linearity: $v_{gs,pk} = \frac{v_{g,pk}}{1 + g_m \times R_{s1}}$ $v_{gs,pk} \leq 0.2 \times V_{OV}$

Gain with source degeneration resistance: $A_{vo} = \frac{-R_D}{1/g_m + R_S}$ (to match the fig below, R_S will be R_{S1})

Common Drain Amplifier Design

$$R_i = R_{G1} \| R_{G2} \qquad R_o = R_S \| 1/g_m \qquad A_{vo} = \frac{R_S}{1/g_m + R_S}$$



STEP 1: Assume $V_{in,pp} = \frac{V_{Load,pp}}{0.85}$ to account for gain less than 1

STEP 2: Pick $V_S \approx \frac{1}{2} V_{DD}$

STEP 3: Find peak load current: $I_{Load,pk} = \frac{V_{Load,pp}/2}{R_{Load}}$

STEP 4: Choose $I_{D,\min}$ large to ensure FET stays ON for different devices

STEP 5: Find worst-case source resistance current: $I_{RS,\max} = I_{D,\min} + I_{Load,pk}$

STEP 6: Find the corresponding minimum voltage: $V_{RS,\min} = V_S - \left(V_{Load,pp}/2\right)$

STEP 7: Find R_S: $R_S = \frac{V_{RS,\min}}{I_{RS,\max}}$

STEP 8: Find the DC drain current: $I_D = \frac{V_S}{R_S}$

STEP 9: Compute $V_{OV} = \sqrt{\frac{2 \times I_D}{K_n}}$

STEP 10: Compute V_G: $V_{GS} = V_{OV} + V_t \qquad V_G = V_{GS} + V_S$

STEP 11: Compute R_{G1} and R_{G2}: $R_{G1} \geq \frac{V_{DD}}{V_G} \times R_{IN} \qquad R_{G2} = \frac{R_{G1}}{\left(\frac{V_{DD}}{V_G} - 1\right)}$

Common Gate Amplifier

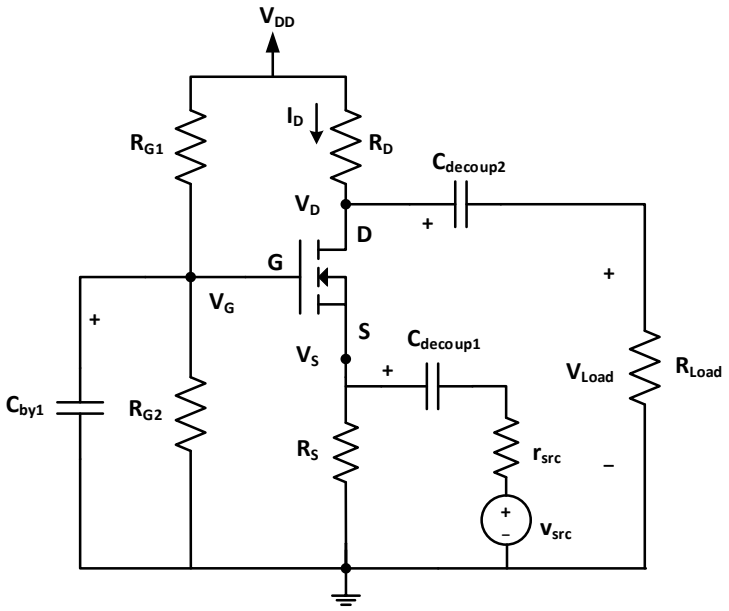
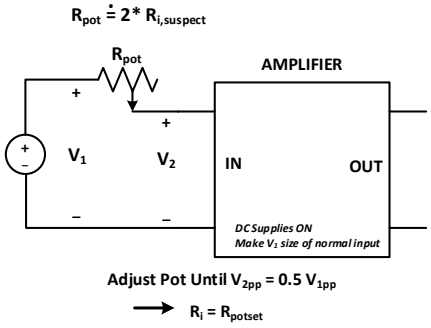


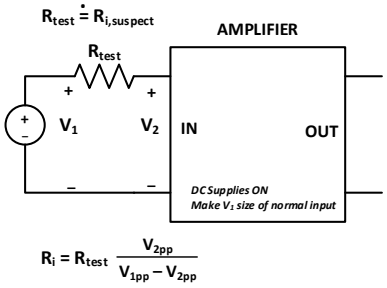
Table of amplifier parameters

	CS no R_S	**CS part bypassed R_S**	**CD**	**CG**
R_i	$R_{G1} \parallel R_{G2}$	$R_{G1} \parallel R_{G2}$	$R_{G1} \parallel R_{G2}$	$R_S \parallel (1/g_m)$
R_o	$R_D \parallel r_o$	$R_D \parallel \left[r_o + (1+g_m r_o) R_{S1} \right] \approx R_D$	$R_S \parallel (1/g_m) \parallel r_o \approx (1/g_m)$	$R_D \parallel \left[r_o (1+g_m R_S) + R_S \right] \approx R_D \parallel r_o$
A_{vo}	$-g_m (R_D \parallel r_o) \approx -g_m R_D$	$\dfrac{-g_m (R_D \parallel r_o)}{1+g_m R_{S1} + \dfrac{R_{S1}+R_D}{r_o}} \approx \dfrac{-R_D}{(1/g_m)+R_{S1}}$	$\dfrac{g_m (R_S \parallel r_o)}{1+g_m (R_S \parallel r_o)} \approx \dfrac{R_S}{R_S + (1/g_m)}$	$\dfrac{g_m r_o}{r_o + R_D} R_D \approx g_m R_D$
A_v	Replace in A_{vo} $R_D \to R_D \parallel R_L$	Replace in A_{vo} $R_D \to R_D \parallel R_L$	Replace in A_{vo} $R_S \to R_S \parallel R_L$	Replace in A_{vo} $R_D \to R_D \parallel R_L$

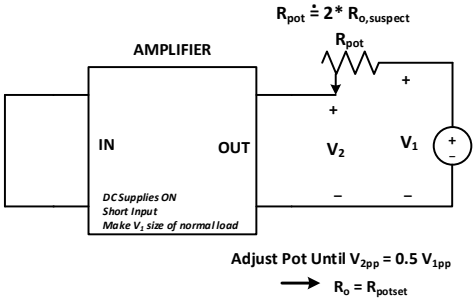
Measuring Amplifier Small-Signal Model Parameters

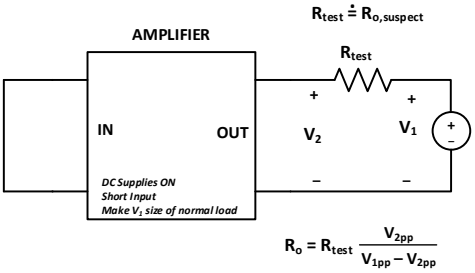
Measuring R_i



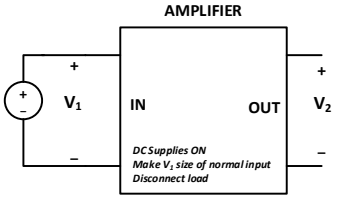


Measuring R_o

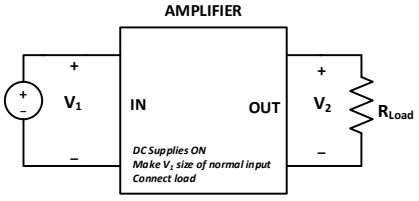




Measuring A_{vo}



Measuring A_v



Power Calculations:

DC Losses: FET: $P_{diss,DC} = V_{DS} \times I_D$ Resistor: $P_{diss,DC} = V^2/R = I^2 * R$

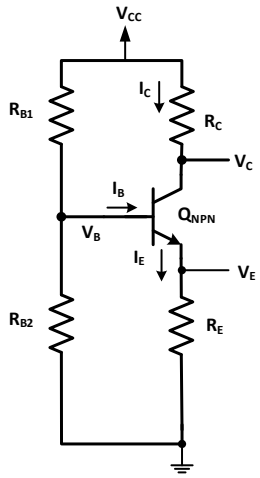
AC Losses: Resistor: $V_{rms} = (V_{pp}/2)/\text{sqrt}(2)$ $P_{diss,AC} = (V_{rms})^2/R$

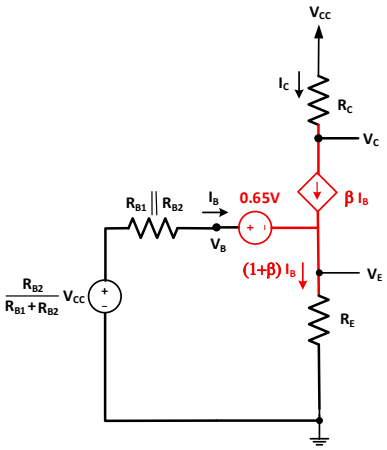
FET: $P_{diss,AC} = 0.5 * v_{ds,pk} * i_{ds,pk}$

Bipolar Junction Transistor (BJT) Material

$$i_B = \frac{i_C}{\beta} \qquad i_E = \frac{i_C}{\alpha} \qquad \alpha = \frac{\beta}{1+\beta} \qquad I_C = I_S e^{V_{BE}/V_T}$$

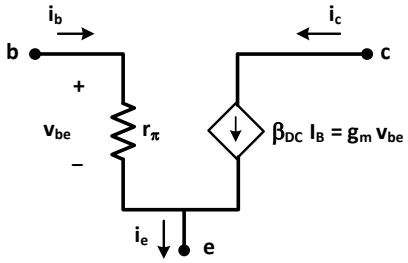
$$I_E = I_B + I_C = (1+\beta) I_B$$





$$I_C = \frac{\beta (V_{th} - V_{BE})}{R_{th} + (1+\beta) R_E} \approx \frac{(V_{th} - V_{BE})}{R_E} \qquad V_{CE} = V_{CC} - I_C R_C - I_E R_E$$

Small-signal Model



$$g_m = \frac{I_C}{V_T} \qquad i_c = g_m v_{be} \qquad r_\pi = \frac{v_{be}}{i_b} = \frac{\beta}{g_m}$$

Biasing Procedure: Given I_C, β, and $V_{BE} = 0.65V$

Step 1: Make $V_{RC} = V_{CE} = 0.4*V_{CC}$ & $V_{RE} = 0.2*V_{CC}$

Step 2: Compute $I_B = I_C/\beta$

Step 3: Choose $I_{RB1} = 50I_B$

Step 4: Find $R_C = V_{RC}/I_C$ $R_E = V_{RE}/(I_C + I_B)$

Step 5: Compute $V_{RB2} = V_{BE} + V_{RE}$

Step 6: Find $I_{RB2} = I_{RB1} - I_B$

Step 7: Compute $R_{B2} = V_{RB2}/I_{RB2}$

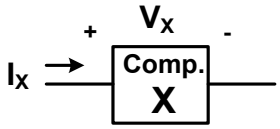
Step 8: Find $I_{RB1} = (V_{CC} - V_{RB2})/I_{RB1}$

ECE245 Helpful Equations Sheet 1

$i = \frac{dq}{dt}$ (amp = coulombs/sec) $v = \frac{dW}{dq} = iR$ (volt = joules/coulomb)

$R = \frac{\rho L}{A} = \frac{v}{i}$ (ohm = volt/amp) $p = \frac{dW}{dt} = vi = \frac{v^2}{R} = i^2 R$ (watt = joules/sec)

$E = \int p dt$ (Energy in joules or kWh)



Passive Sign Convention $P_X = V_X I_X$ absorbed by "Comp. X"

Kirchhoff's Laws: KVL: $\sum_{Loop} V = 0$ KCL: $\sum_{node} I = 0$

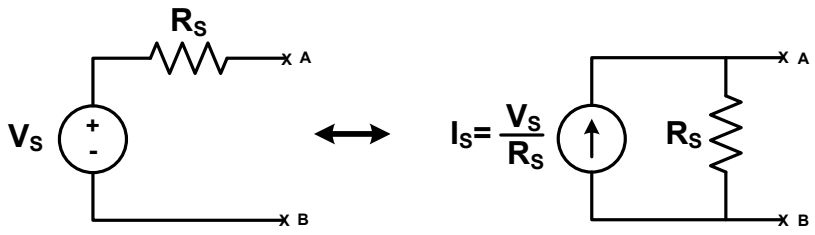
Resistor Comb.: **SERIES** **PARALLEL**

$$R_{eq} = R_1 + R_2 + R_3 + \ldots \quad R_{eq} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots} \qquad R_{2inParallel} = \frac{R_1 R_2}{R_1 + R_2}$$

Voltage Divider: $V_X = \frac{R_X}{\sum_{series} R} \times V_{series}$ **Current Divider**: $I_X = \frac{R_{eq,parallel}}{R_X} \times I_{TOT,par}$

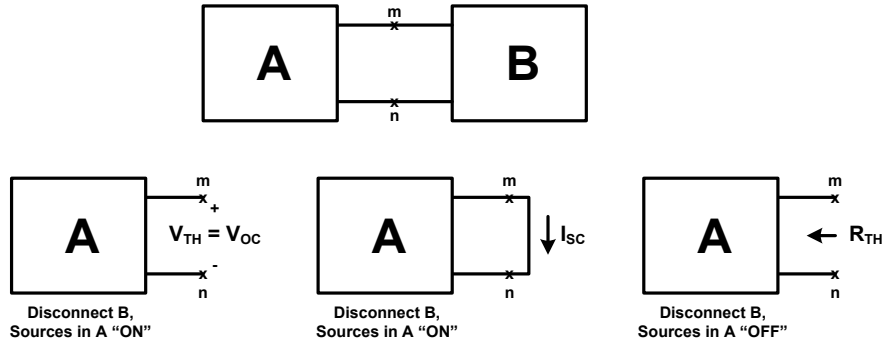
Nodal Analysis: (KCL) $I_{out,X} = \frac{V_X - V_Y}{R_{XY}}$ **Mesh Analysis**: (KVL) $V_{12} = (I_1 - I_2) R_{12}$

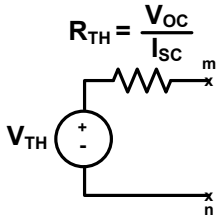
Source Transformation:



Superposition: $V_X = V_{X1} + V_{X2} + V_{X3} + \ldots = V_X\big|_{\substack{S1,ON \\ S2,S3\ldots OFF}} + V_X\big|_{\substack{S2,ON \\ S1,S3\ldots OFF}} + V_X\big|_{\substack{S3,ON \\ S1,S2\ldots OFF}} + \ldots$

Thevenin Equivalent: Given Networks A & B, the Thevenin of Network A is found from





Maximum Power Transfer: Set $R_{Load} = R_{out} = R_{TH}$ $\quad P_{Load} = I_{Load}^2 R_{Load} = \frac{V_{Load}^2}{R_{Load}}$

Metric Prefixes

Giga	G	10^9
Mega	M	10^6
Kilo	k	10^3
Milli	m	10^{-3}
Micro	μ	10^{-6}
Nano	n	10^{-9}
Pico	p	10^{-12}

Resistor Color Code **4th Band**: Gold = 5%; Silver = 10%; None = 20%

Black	Brown	Red	Orange	Yellow	Green	Blue	Violet	Gray	White
0	1	2	3	4	5	6	7	8	9

Resistor Wattage/Size

1/8W	1/4W	1/2W	1W	2W
3.3mm	6.0mm	8.5mm	11mm	15mm